Course

Test II

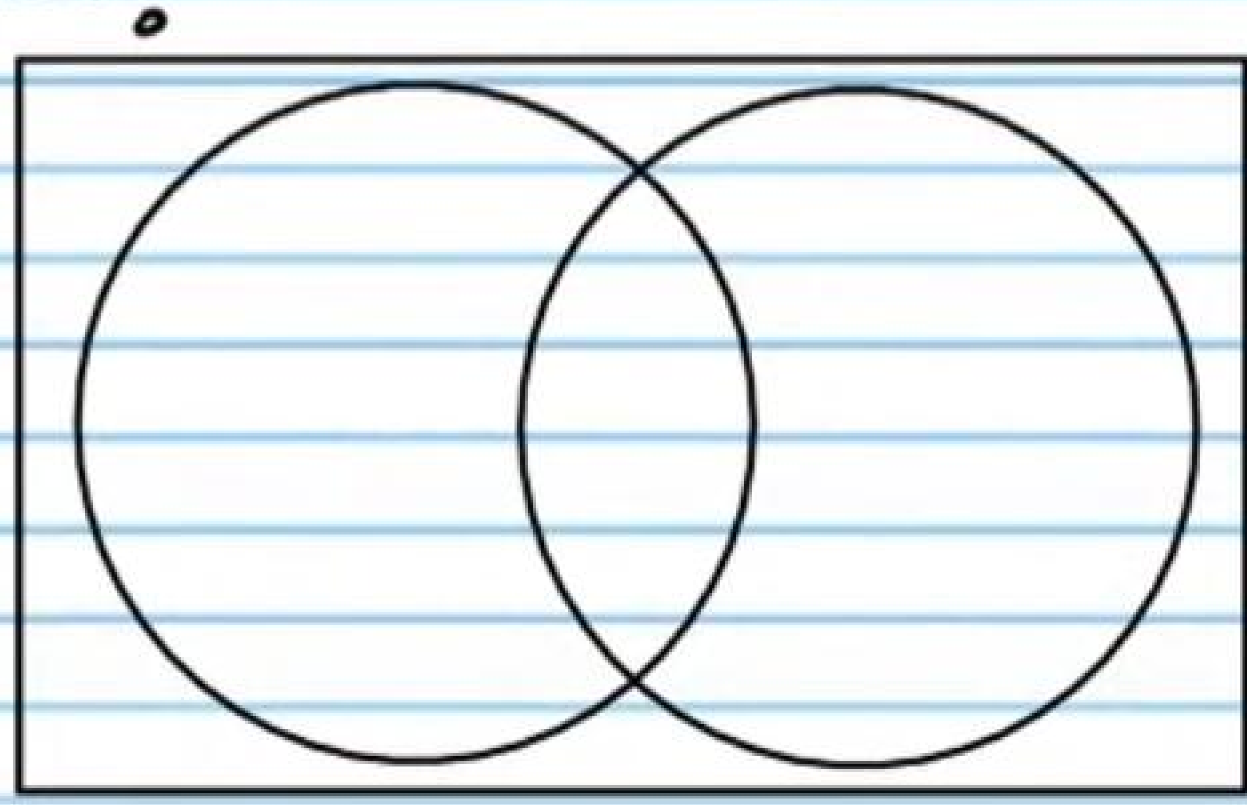
I) Write your name, the date, and the course title (i.e. Precalculus).
II) Show all work.
III) Draw a rectangle around your final answer for each problem.
IV) You may use a calculator except for graphing.
V) Each problem is worth 4 points for a total of 100 points.

① 72 children are served sandwiches for a picnic lunch. 50 students want peanut butter on their sandwich. 43 students want jelly on their sandwich. 35 students want both peanut butter and jelly on their sandwich.

i) How many children want peanut butter on their sandwich but not jelly?

ii) How many children want jelly on their sandwich but not peanut butter?

iii) How many children will have either peanut butter or jelly on their sandwich?



iv) How many children want neither peanut butter nor jelly on their sandwich?

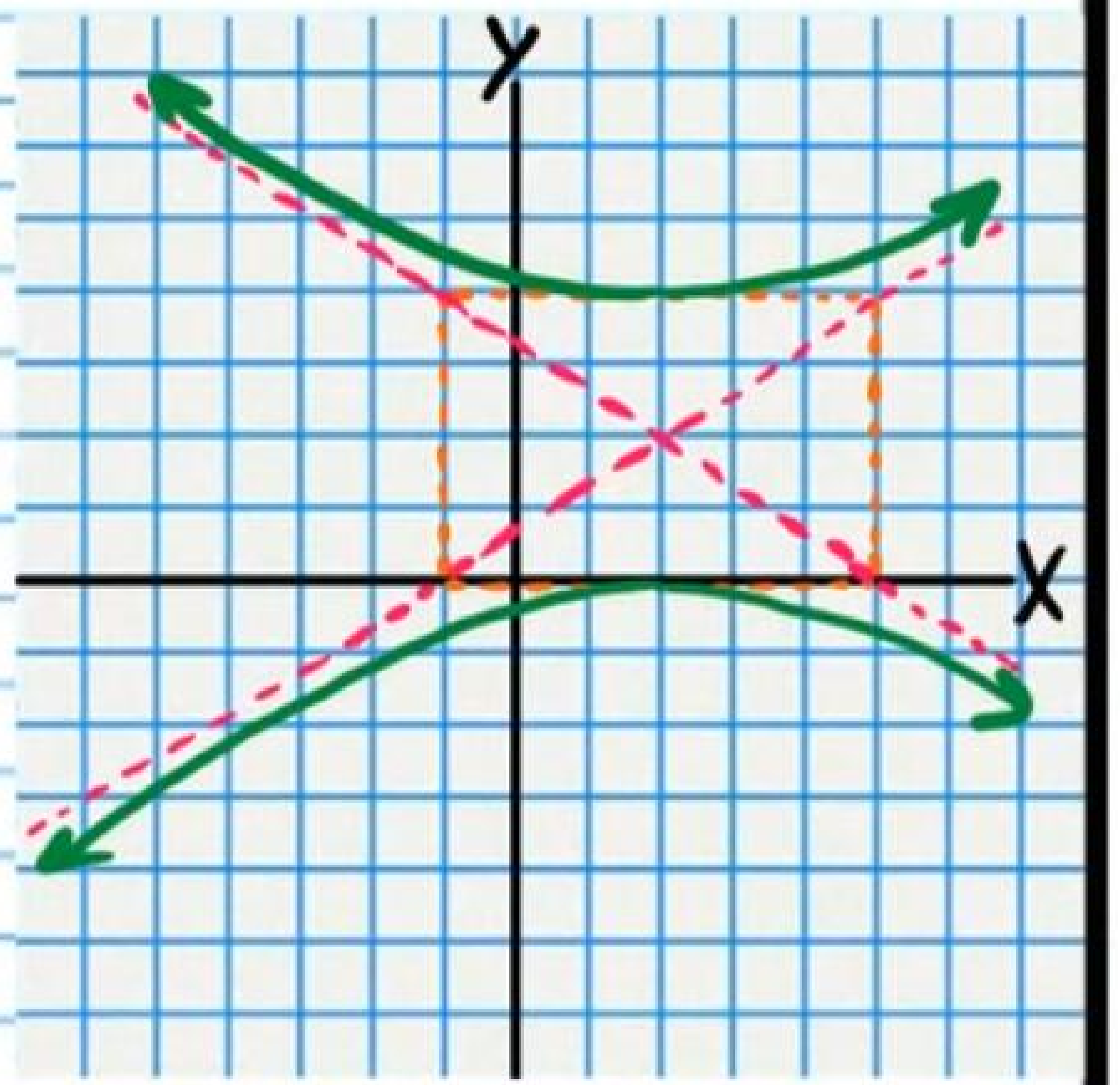
v) What is the probability that a randomly chosen child will want jelly on his sandwich?

vi) What is the probability that a randomly chosen child will want either peanut butter or jelly on his sandwich?

vii) What is the probability that a randomly chosen child will want neither peanut butter nor jelly on his sandwich?

viii) What is the probability that a randomly chosen child will want both peanut butter and jelly on his sandwich?

② Write an equation of the graph.



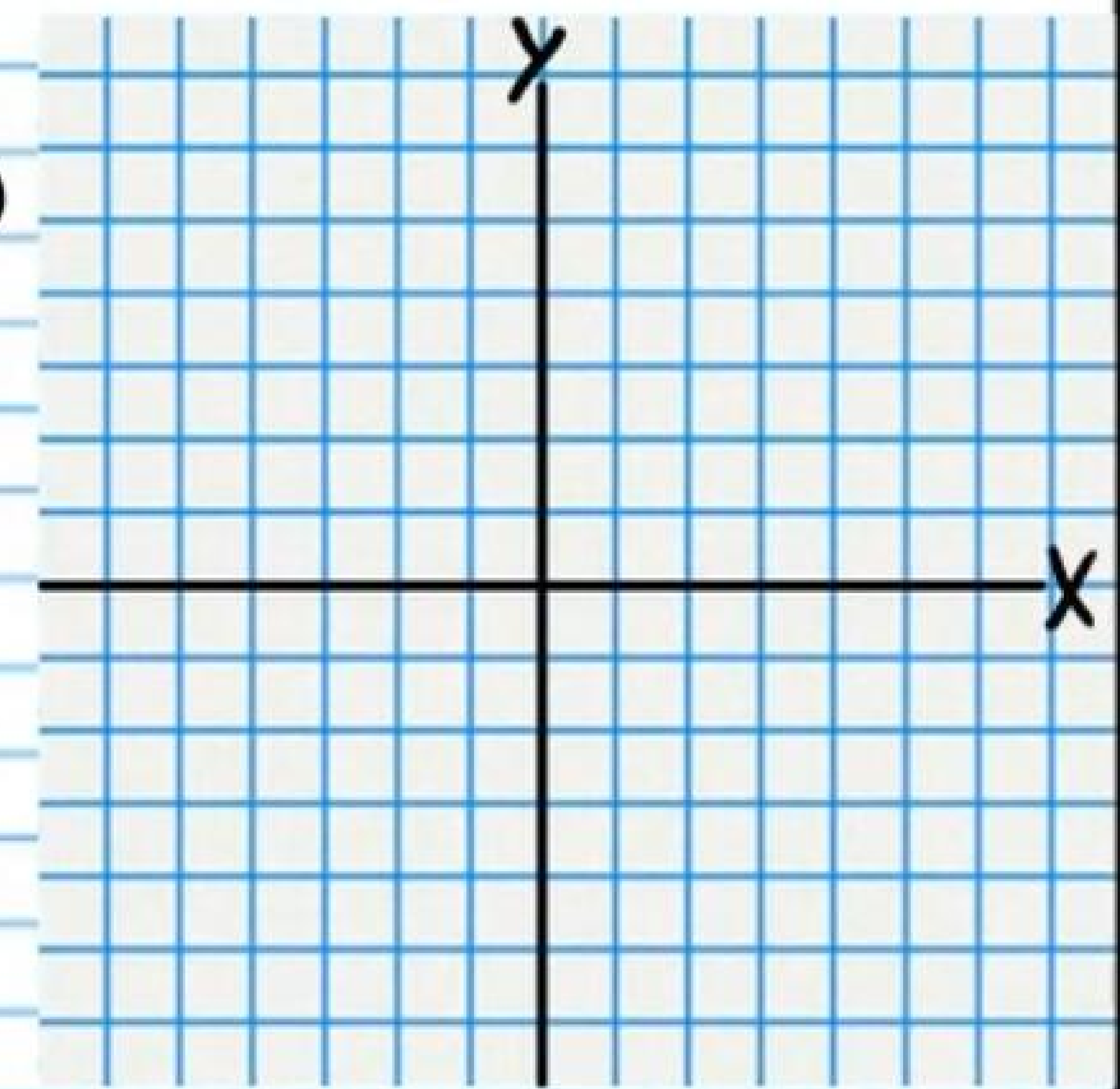
③ Use the Binomial Theorem to expand the following expression.

$(4x-5y)^3$

④ The probability that a bus will not arrive on time is 24%. What is the probability that this bus will arrive on time?

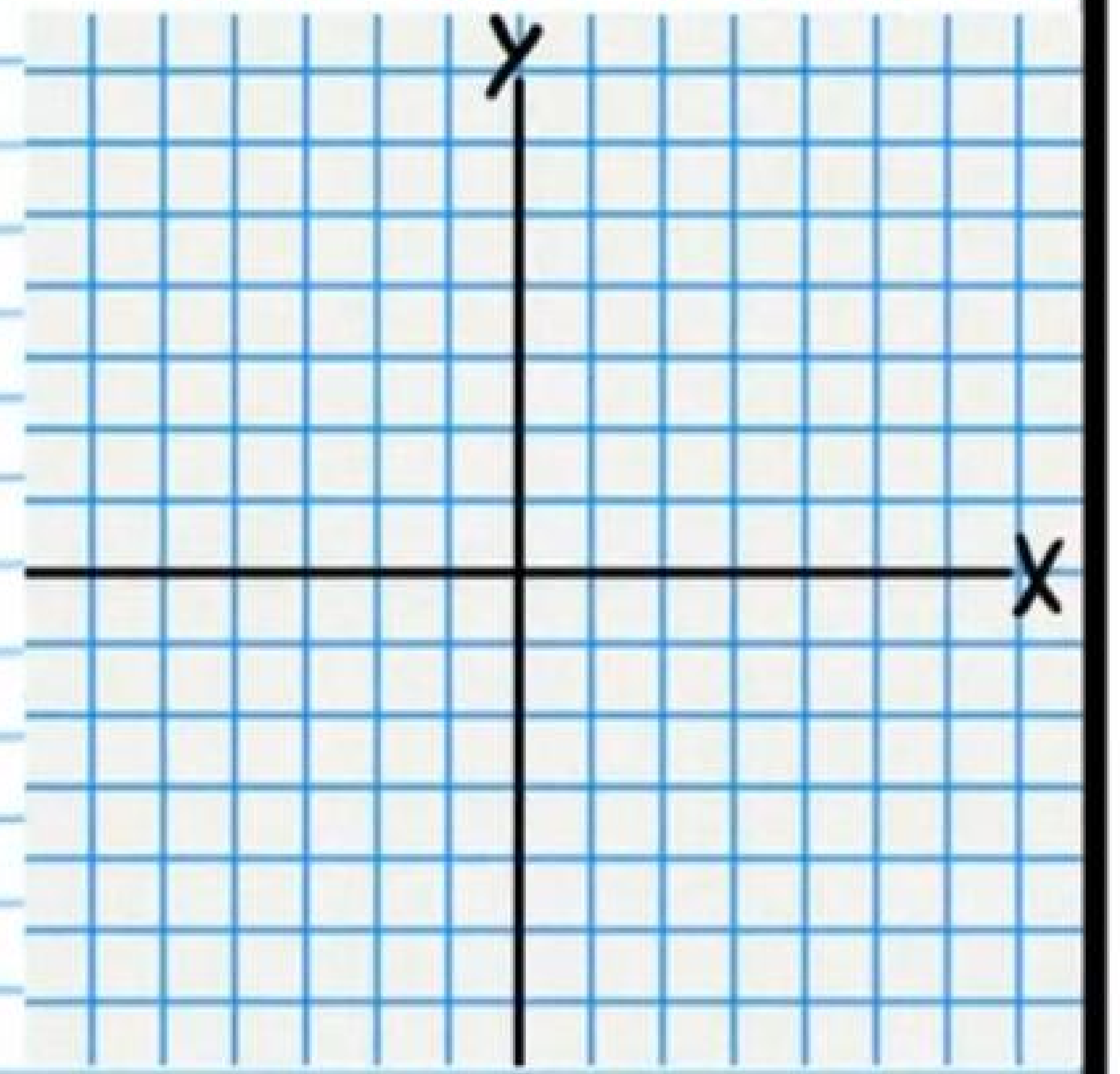
⑤ Graph the following equation. ★ You must first convert to a recognizable form.

$9x^2-10y^2+72x-20y+44=0$



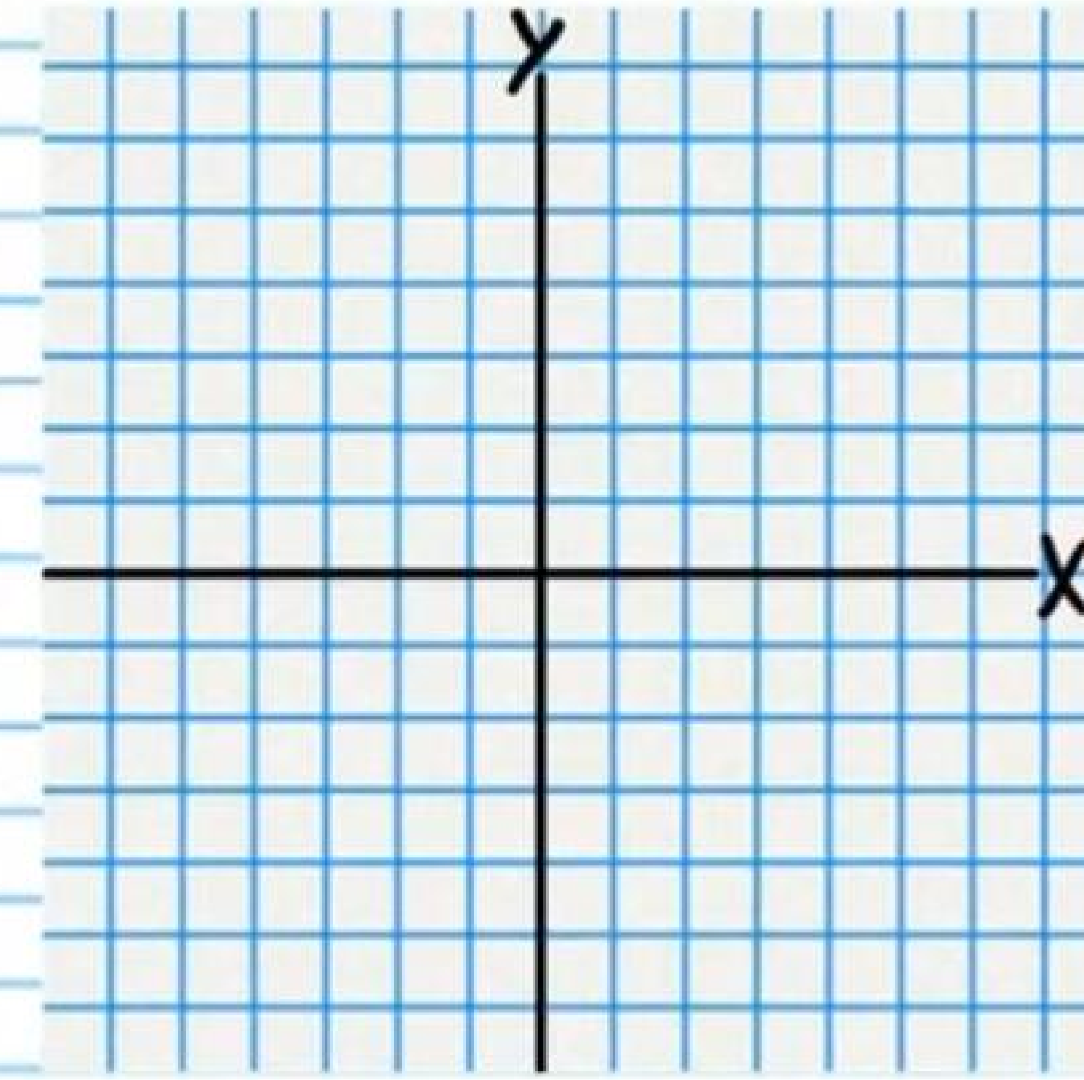
⑥ Graph the following equation. ✱ You must first convert to a recognizable form.

$x^2+4y^2-6x+16y+21=0$



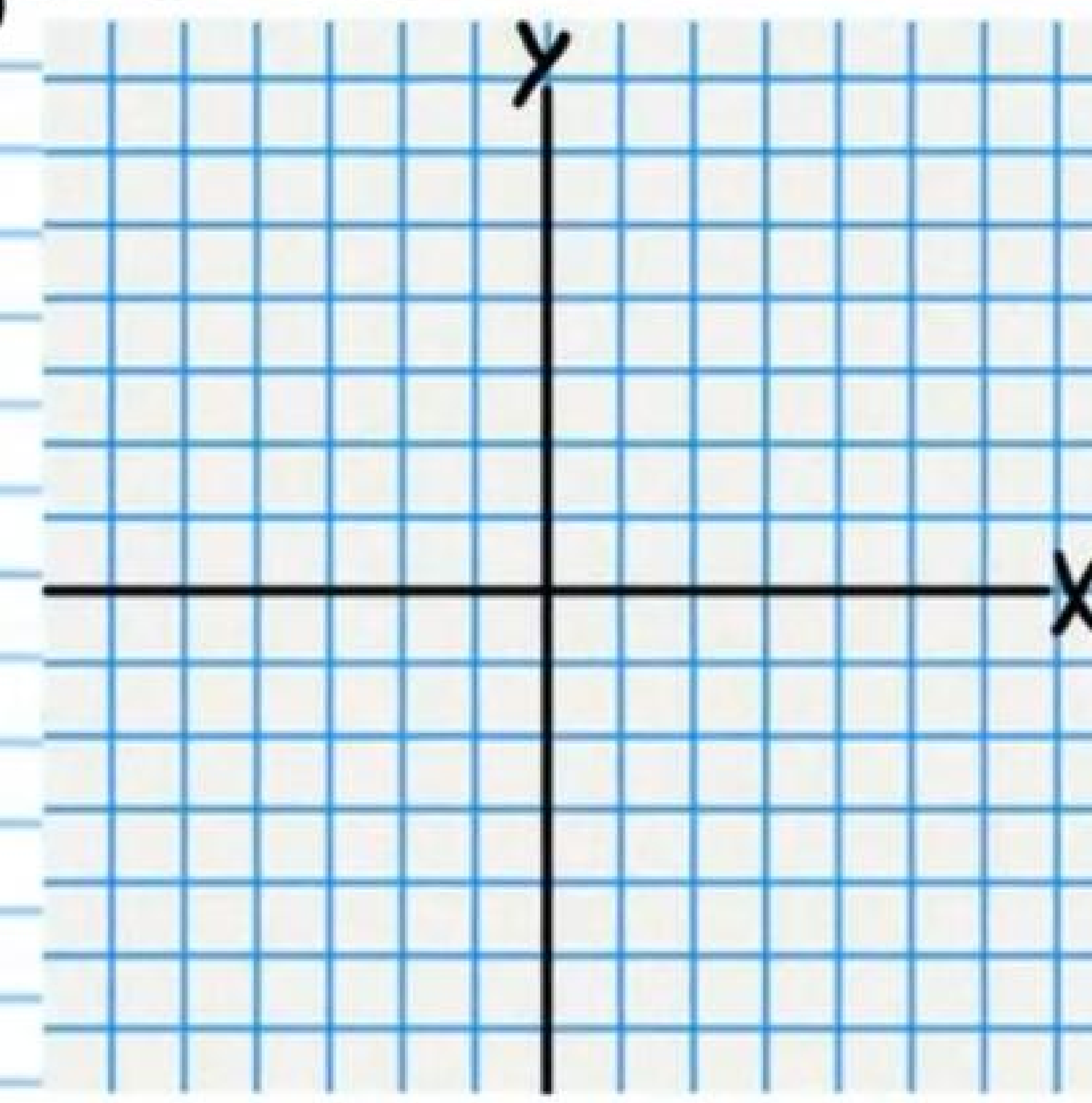
⑦ Graph the following equation. Write the coordinates of the focus and the equation of the directrix.

$y+3=\frac{1}{2}(x-2)^2$



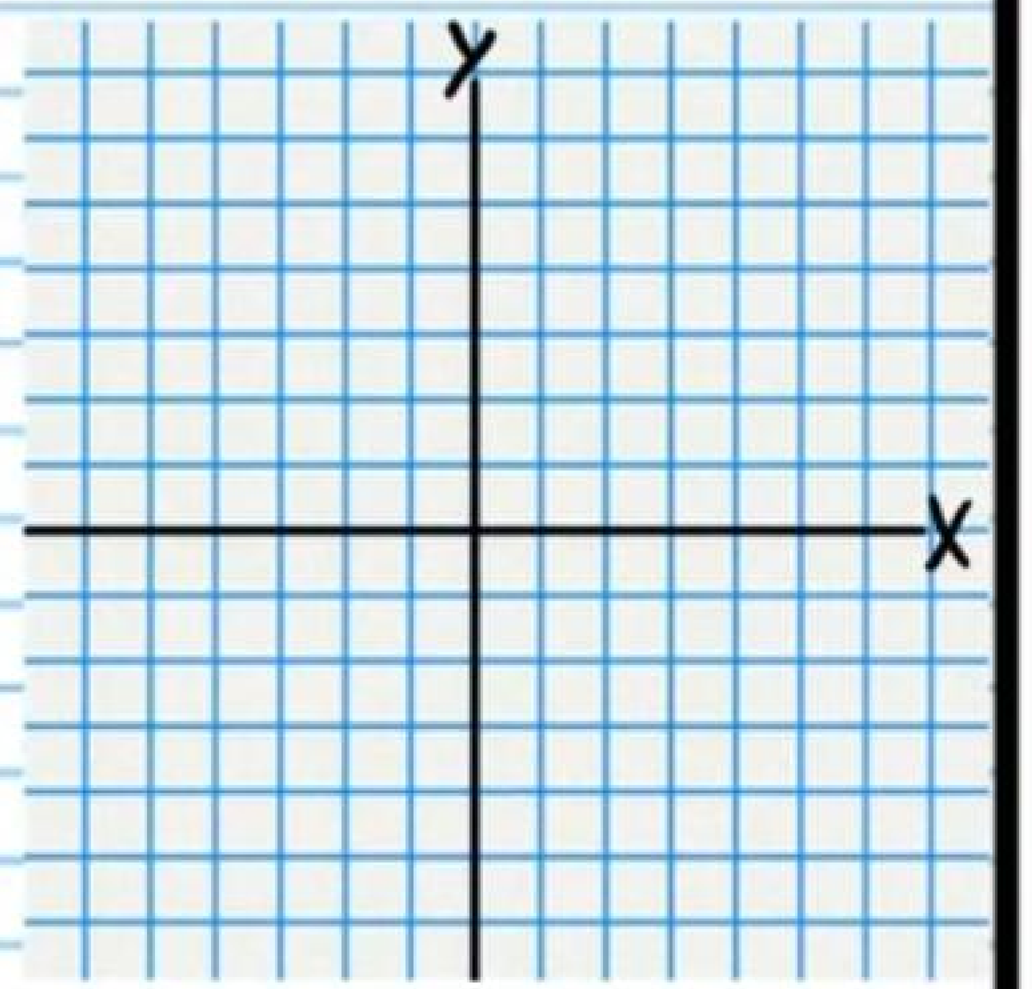
⑧ Graph the following equation. Write the coordinates of the center and the length of the radius.

$\frac{(x+3)^2}{4}+\frac{y^2}{4}=1$



⑨ Graph the following equation. Write the coordinates of the foci, draw the asymptotes as dotted lines, and write their equations.

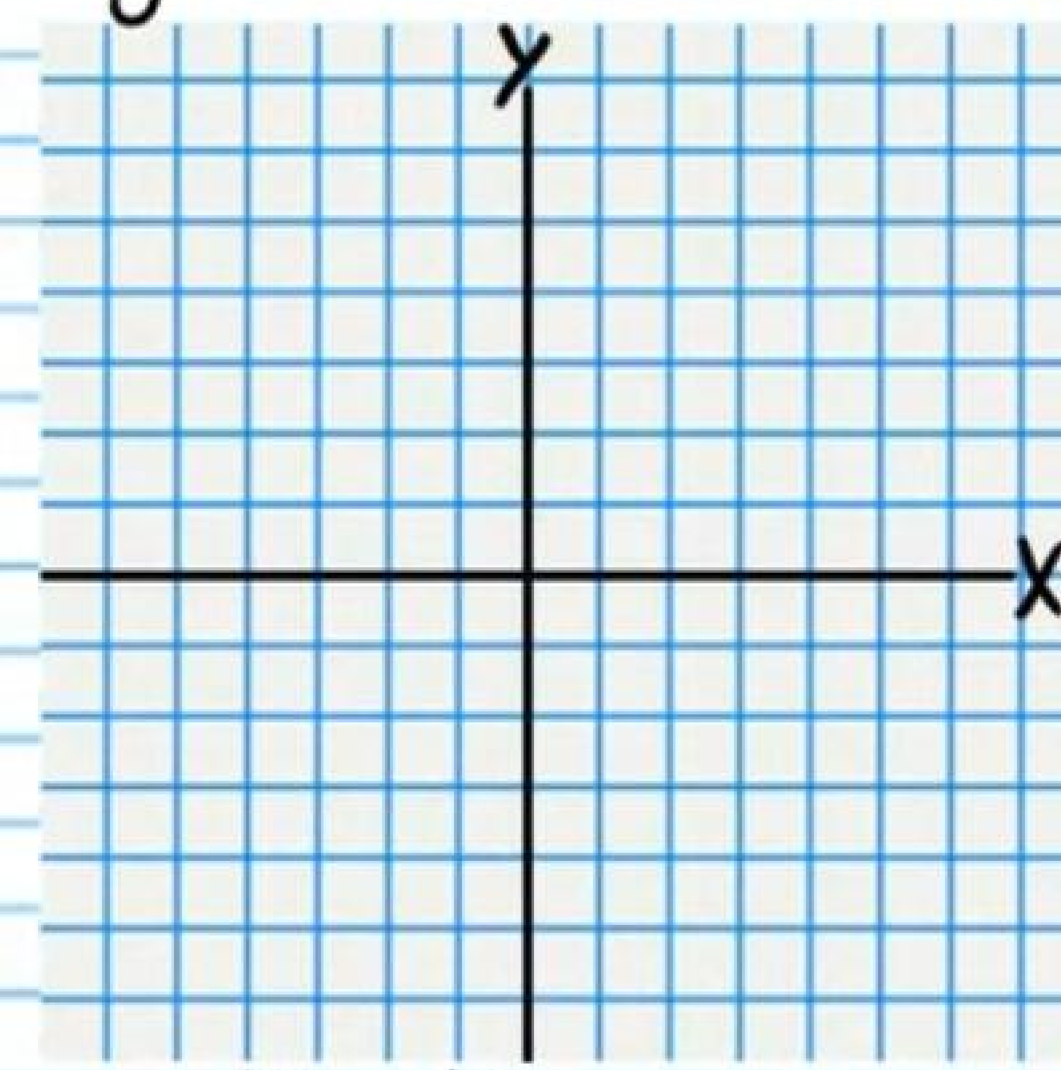
$$\frac{(y-1)^2}{25} - \frac{(x+3)^2}{4} = 1$$



Asymptotes

⑩ Find an equation of the ellipse with the following characteristics and graph the equation.

Vertices at $(-4,-1)$ and $(2,-1)$, length of minor axis is 4.



⑪ If the 5^{th} term of an arithmetic sequence is 83, and the 10^{th} term is 78, what is the 6^{th} term?

⑫ Find the coefficient of the b^4 term in the expansion of $(b-1)^{10}$?

⑬ A person is chosen at random from a group attending a conference. The probability that the person visited Spain is $\frac{1}{5}$. The probability that the person visited Japan is $\frac{2}{9}$. The probability that the person visited both Spain and Japan is $\frac{1}{15}$. What is the probability that this randomly chosen person visited either Spain or Japan?

⑭ Find each sum.

i) $\sum_{n=0}^{4} 1-n^2$

ii) $\sum_{n=1}^{\infty} \left(-\frac{3}{8}\right)^n$

⑮ Write an equation of the n^{th} term of the sequence below in recursive form. Also, show the value of the first term.

$\frac{1}{30}, -\frac{1}{10}, \frac{3}{10}, -\frac{9}{10} \ldots$

(16) Find the sum.

$$\sum_{n=3}^{90} -\frac{1}{3}n+2$$

(17) Find the sum of the first 20 terms in the geometric sequence below. You may leave your answer in exponential form if parts are very large or very small.

$\frac{3}{2000}, \frac{3}{1000}, \frac{3}{500}, \ldots$

For problem #'s 18–19, determine if each sequence is arithmetic, geometric, or neither. Then find an expression for the n^{th} term. Assume that the first term is produced when $n=1$.

(18) i) $4, 9, 16, 25, \ldots$ ii) $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$

(19) i) $-0.1, 0.1, 0.3, \ldots$ ii) $-9, 81, -729, \ldots$

(20) Investors must choose a 4-person management team. The positions are president, vice president, product manager, and sales manager. If the investors can choose from 9 employees of the company, how many outcome variations are possible? Note: A team where Douglas is the president and Marvin is the vice president is different from a team where Marvin is the president and Douglas is the vice president.

(21) Jerry is preparing to show his coin collection to his class. He does not want to bring all of his coins, so he will choose some out of his collection. He has 6 distinct dimes, 6 distinct pennies, and 7 distinct quarters. If he decides to bring 2 pennies, 5 dimes, and 3 quarters, how many possible outcome variations are there for his coin presentation? Although some coins have the same denomination (e.g. 4 coins are all pennies), every coin is unique in some way (e.g. One quarter might be made of silver while another quarter could be made of a copper alloy).

(22) There are 9 blue cards, 5 purple cards, and 4 orange cards in a deck. 3 cards are chosen at random.

i) What is the probability that all 3 are orange?

ii) What is the probability that 2 cards will be blue and 1 will be orange?

iii) What is the probability that 2 will be purple and 1 will be blue?

iv) What is the probability that 1 card will be purple, 1 will be blue, and 1 will be orange?

(23) i) Find the 81^{st} term in the arithmetic sequence below.

8, 18, 28, ...

ii) Find the 53^{rd} term in the geometric sequence below.
You may leave your answer in exponential form.

0.7, 2.1, 6.3, ...

(24) A person is selected at random. The probability that the person will be between 29 and 39 years old is 0.17. The probability that the person will be between 45 and 55 years old is 0.15. What is the probability that a randomly selected person will be either between 29 and 39 years old or between 45 and 55 years old?

(25) There are 20 different types of amino acids used by biological organisms in nature. Amino acids are connected in strands called polypeptides. How many polypeptide variations are possible for a strand that has 7 amino acids? Note: Repetition is allowed.

Answers

① i) $\boxed{\text{15 children}}$ ii) $\boxed{\text{8 children}}$ iii) $\boxed{\text{58 children}}$

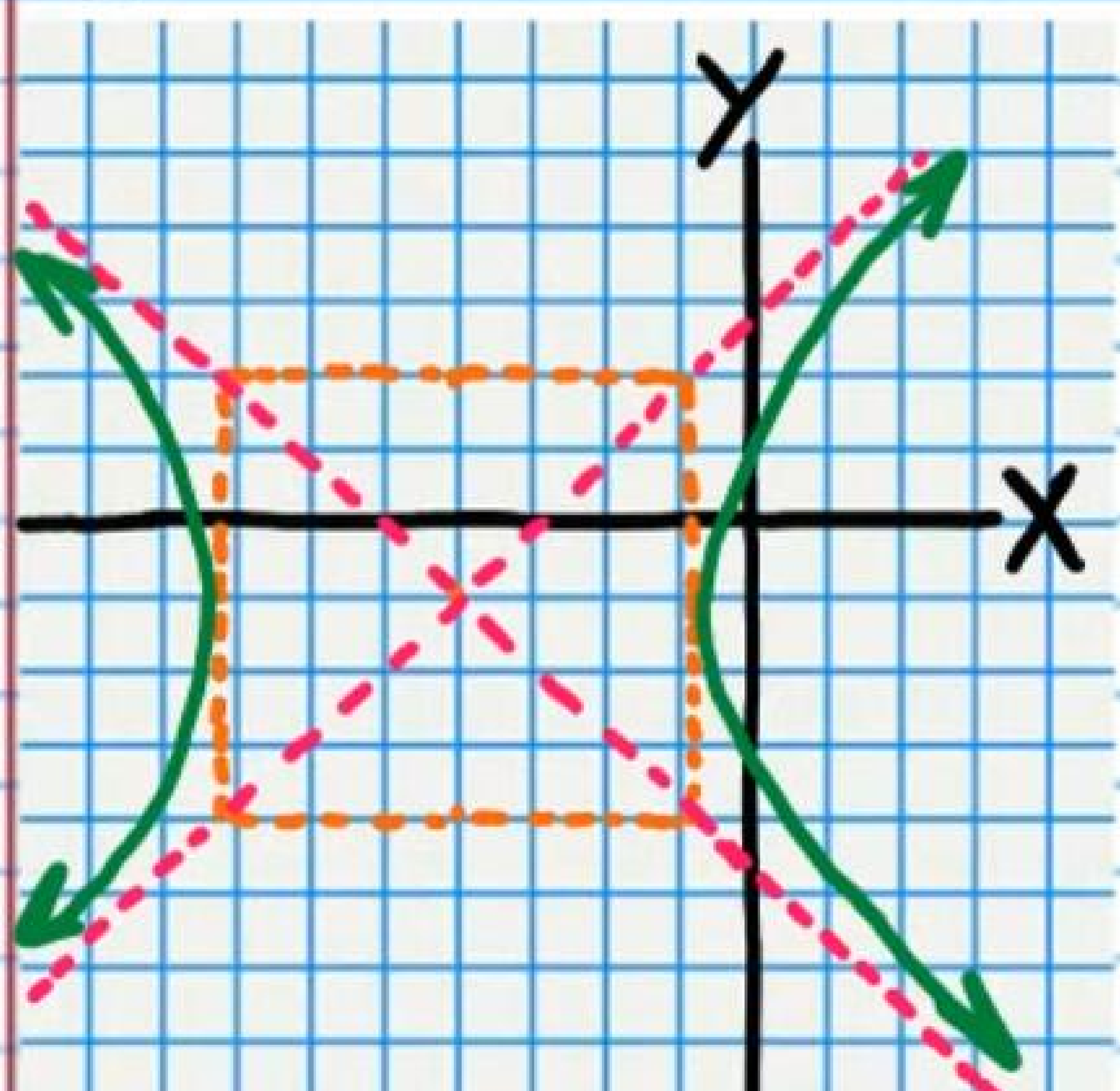
iv) $\boxed{\text{14 children}}$ v) $\boxed{\frac{43}{72}}$ vi) $\boxed{\frac{29}{36}}$ vii) $\boxed{\frac{7}{36}}$ viii) $\boxed{\frac{35}{72}}$

② $\boxed{\frac{(y-2)^2}{4}-\frac{(x-2)^2}{9}=1}$

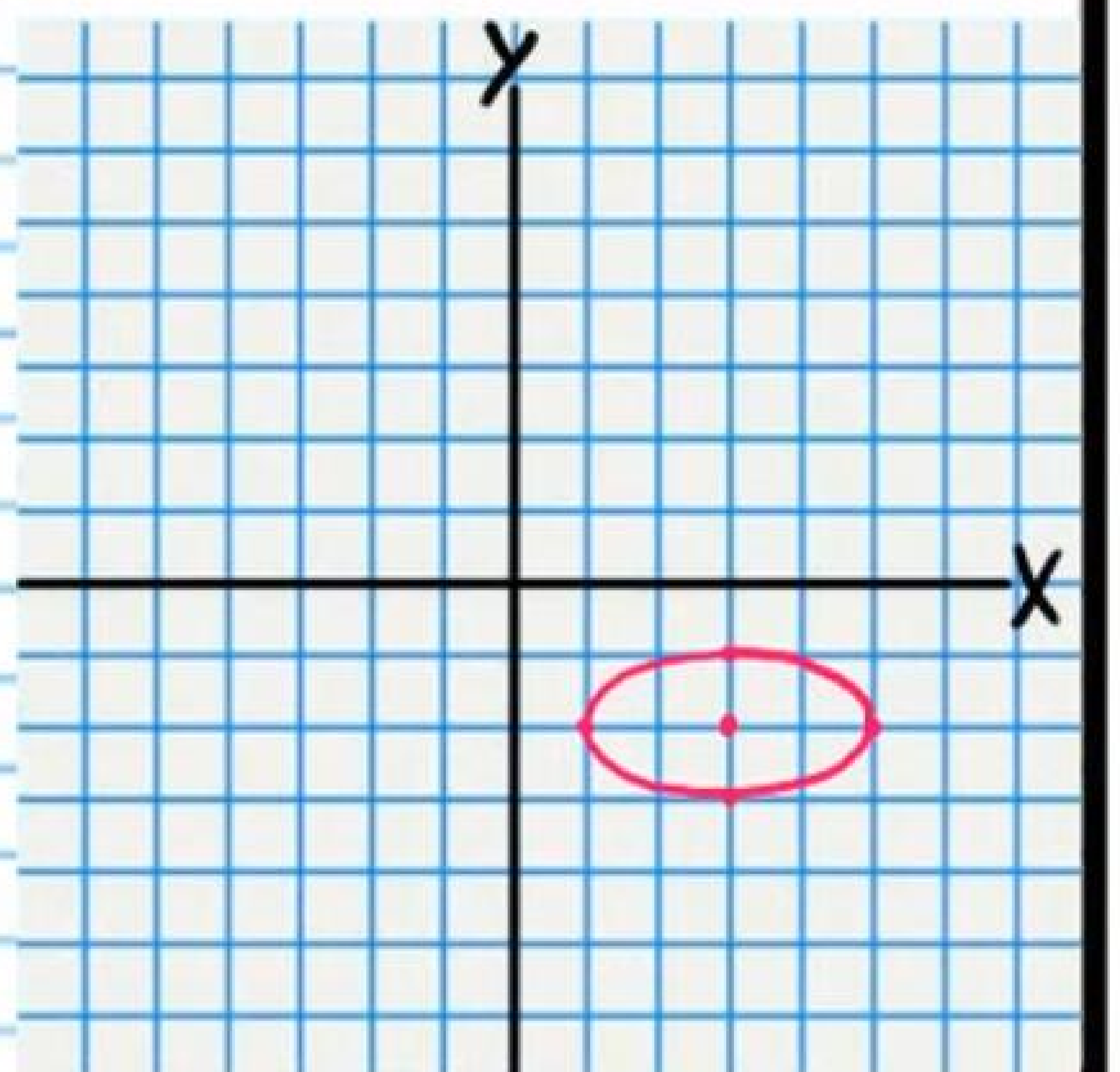
③ $\boxed{64x^3-240x^2y+300xy^2-125y^3}$

④ $\boxed{76\%}$

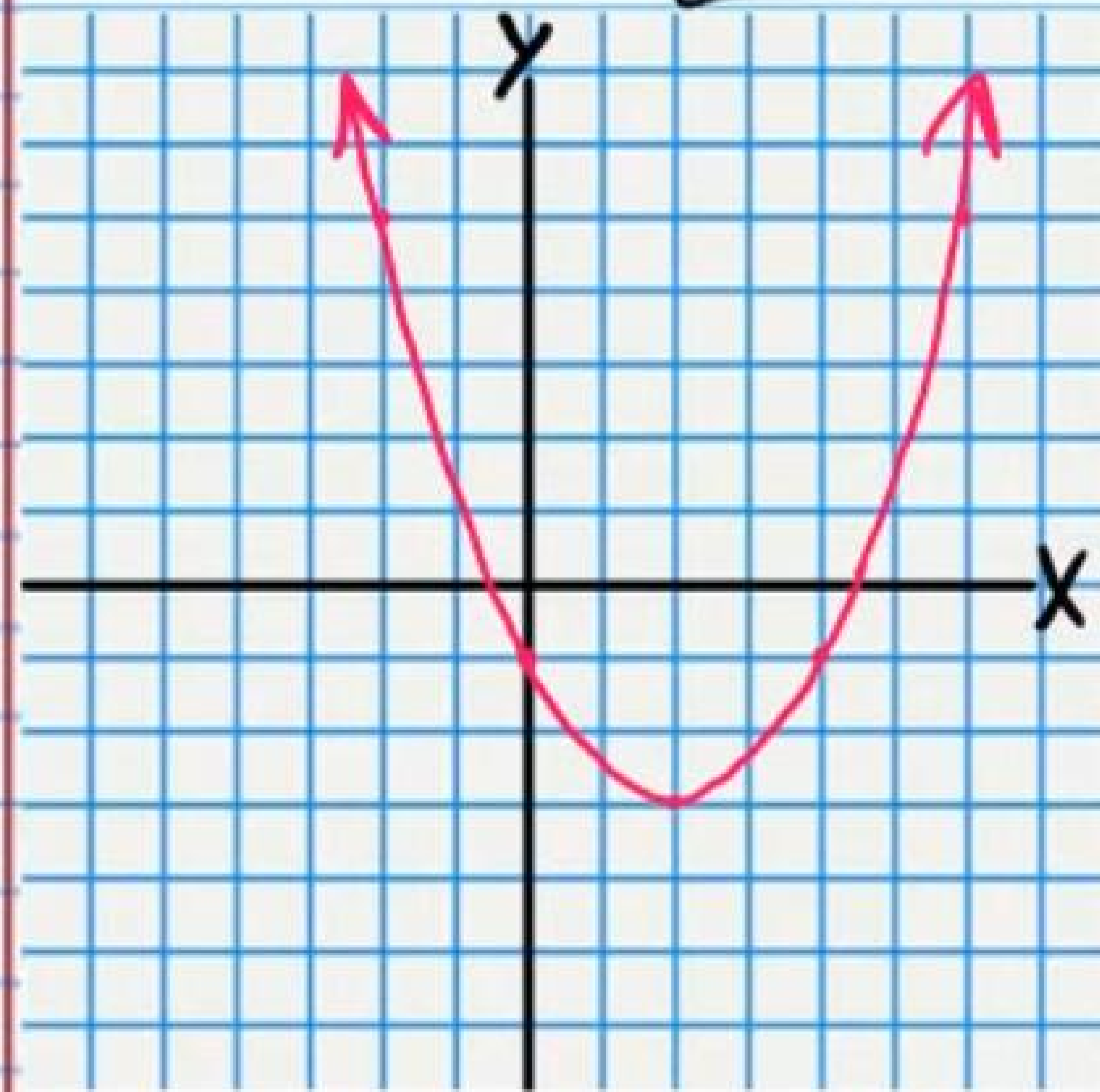
⑤



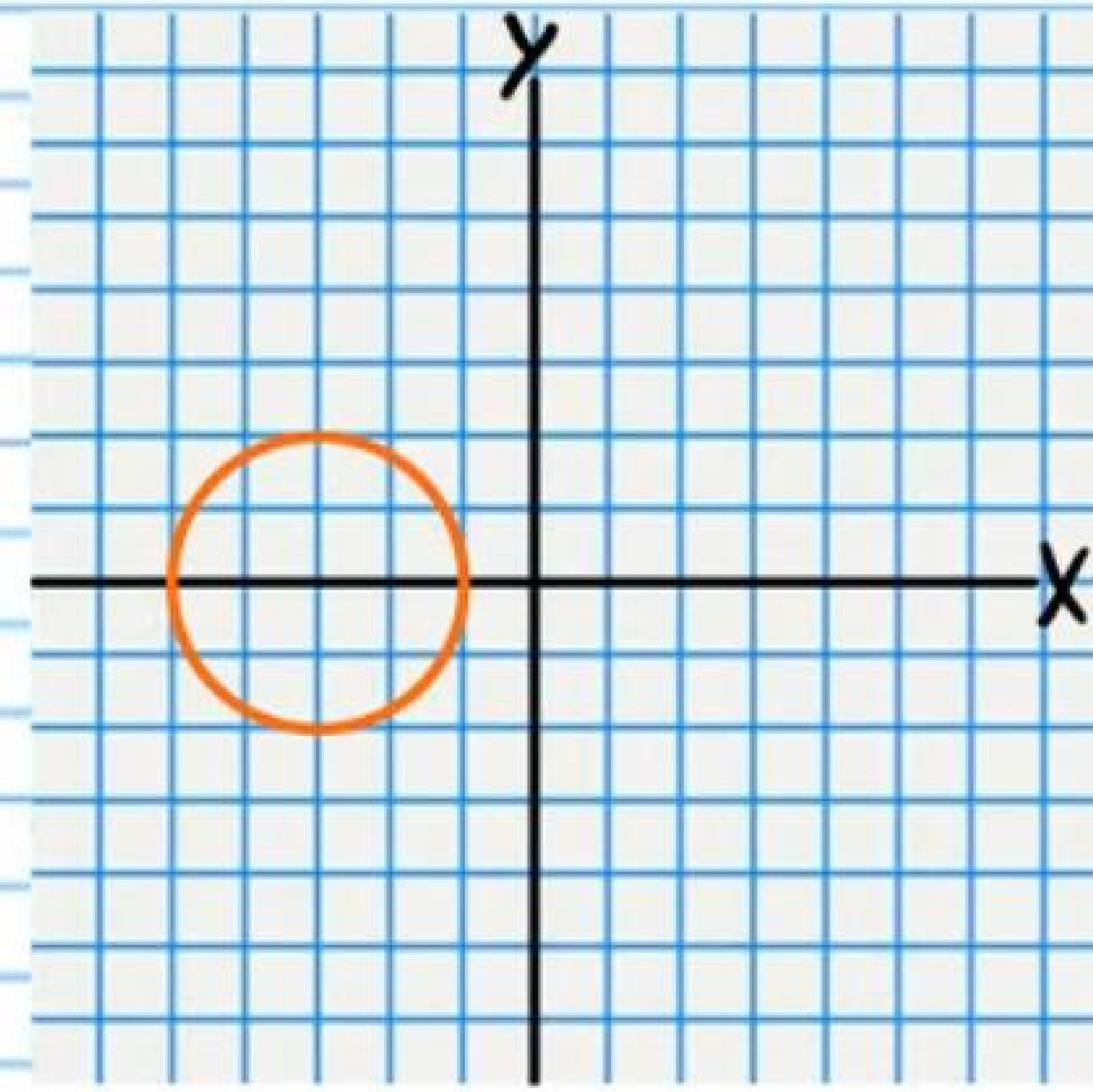
⑥



⑦ $\boxed{F(2,-2\tfrac{1}{2})}$ $\boxed{y=-3\tfrac{1}{2}}$



⑧ $\boxed{C(-3,0)}$ $\boxed{r=2}$

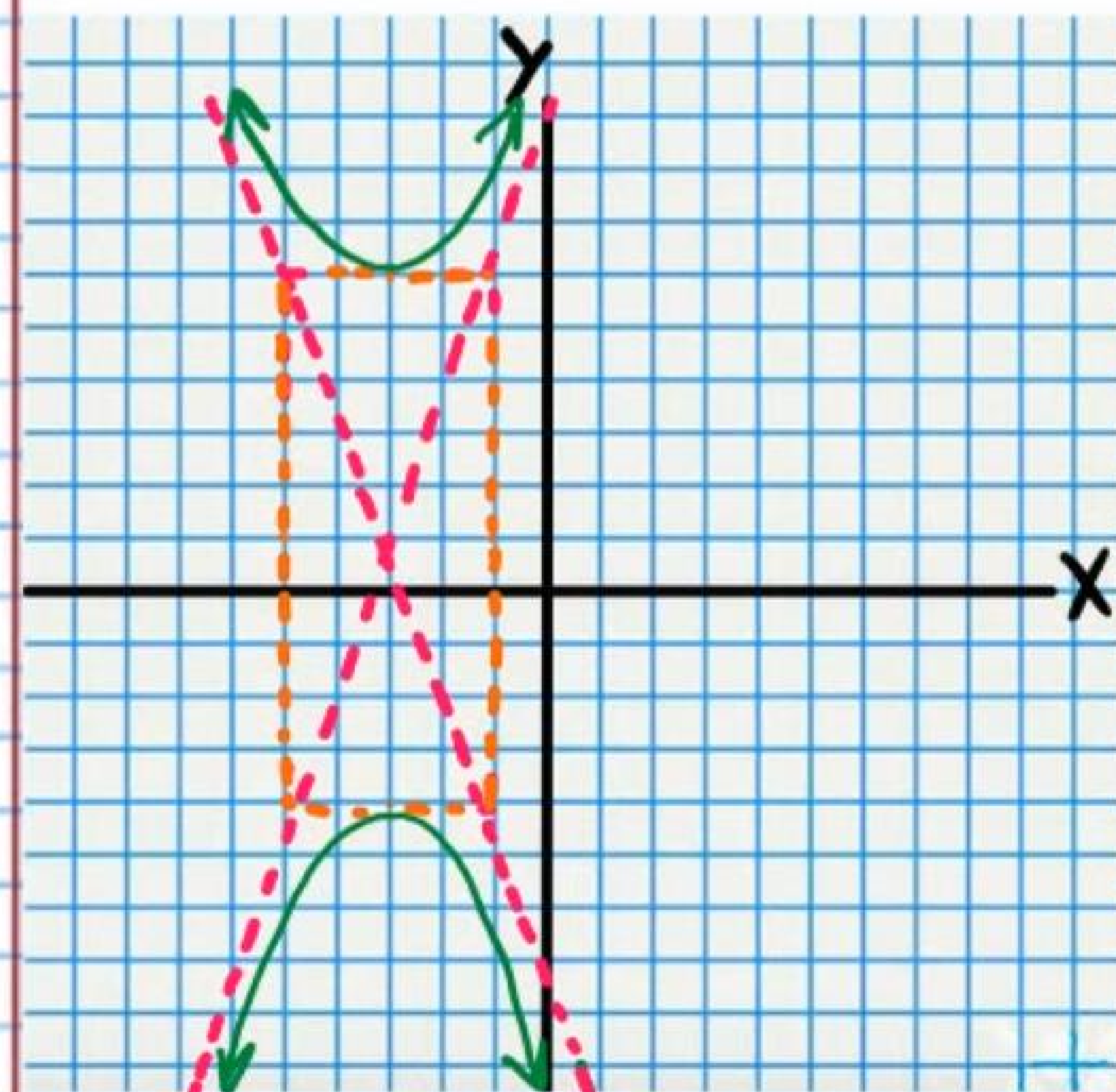


⑨ $\boxed{F_1(-3,1+\sqrt{29})}$, $\boxed{F_2(-3,1-\sqrt{29})}$

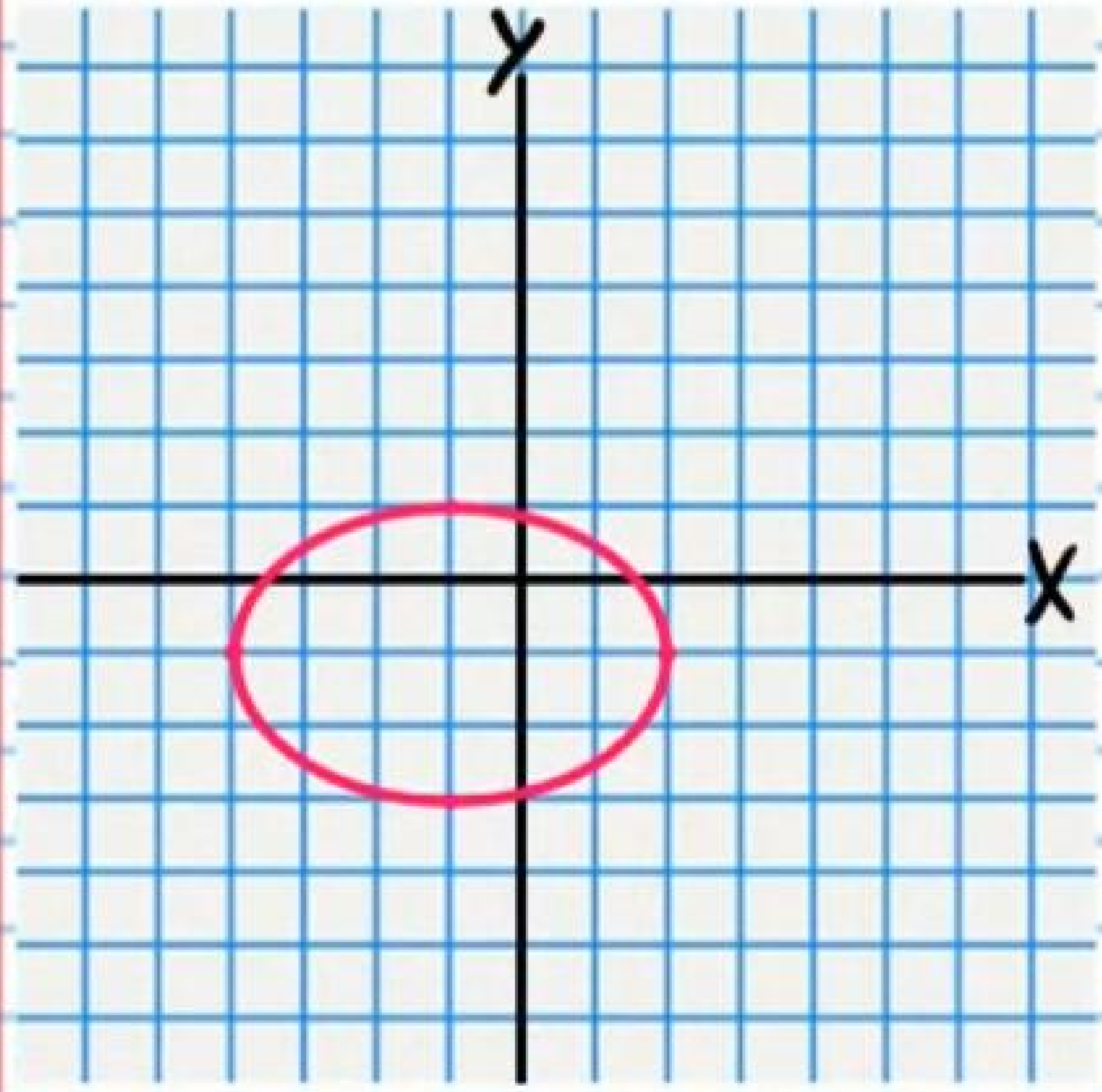
Asymptotes

$\boxed{y-1=\frac{5}{2}(x+3)}$

$\boxed{y-1=-\frac{5}{2}(x+3)}$



(10) $\boxed{\dfrac{(x+1)^2}{9} + \dfrac{(y+1)^2}{4} = 1}$



(11) $\boxed{82}$

(12) $\boxed{210}$

(13) $\boxed{\dfrac{16}{45}}$

(14) i) $\boxed{-25}$ ii) $\boxed{-\dfrac{3}{11}}$

(15) $\boxed{\left\{a_1 = \dfrac{1}{30},\ a_n = -3a_{n-1}\right\}}$

(16) $\boxed{-1{,}188}$

(17) $\boxed{\dfrac{3}{2000}(2^{20}-1)}$ or $\boxed{\dfrac{125{,}829}{80}}$

(18) i) $\boxed{\text{Neither}}$

$\boxed{a_n = n^2}$ or $\boxed{\{n^2\}}$

ii) $\boxed{\text{Neither}}$

$\boxed{a_n = \dfrac{1}{n+2}}$ or $\boxed{\left\{\dfrac{1}{n+2}\right\}}$

(19) i) $\boxed{\text{Arithmetic}}$

$\boxed{a_n = 0.2n - 0.3}$ or $\boxed{\{0.2n - 0.3\}}$

ii) $\boxed{\text{Geometric}}$

$\boxed{a_n = -9(-9)^{n-1}}$ or $\boxed{a_n = (-9)^n}$ or $\boxed{\{-9(-9)^{n-1}\}}$

or $\boxed{\{(-9)^n\}}$

(20) $\boxed{\text{3,024 outcome variations}}$

(21) $\boxed{\text{3,150 outcome variations}}$

(22) i) $\boxed{\dfrac{1}{204}}$ ii) $\boxed{\dfrac{3}{17}}$ iii) $\boxed{\dfrac{15}{136}}$ iv) $\boxed{\dfrac{15}{68}}$

(23) i) $\boxed{808}$ ii) $\boxed{0.7\,\dfrac{1-3^{53}}{-2}}$ or $\boxed{-\dfrac{7}{20}(1-3^{53})}$

(24) $\boxed{0.32}$ or $\boxed{32\%}$ or $\boxed{\dfrac{8}{25}}$

(25) $\boxed{20^7}$ or $\boxed{\text{1,280,000,000 variations}}$